

Unlearning Backdoor Attacks in Federated Learning

Chen Wu*, Sencun Zhu*

Computer Science and Engineering
Penn State University

University Park, PA, United States
cvw5218@psu.edu, sxz16@psu.edu

Prasenjit Mitra†

Information Science & Technology
Penn State University

University Park, PA, United States
pum10@psu.edu

Wei Wang‡

Department of Information Security
Beijing Jiaotong University

Beijing, China
wangwei1@bjtu.edu.cn

Abstract—Federated learning systems are constantly under the looming threat of backdoor attacks. Despite significant progress in mitigating such attacks, the challenge of effectively removing a potential attacker’s influence from the trained global model remains unresolved. In this paper, we present a novel federated unlearning method that is suitable for backdoor removal. By leveraging historical updates subtraction and knowledge distillation, our approach can maintain the models’s performance while completely removing the backdoors implanted by the attacker from the model. It can be seamlessly applied to various types of neural networks and does not require clients’ participation in the unlearning process. Through experiments on diverse computer vision and natural language processing datasets, we demonstrate the effectiveness and efficiency of our proposed method. The promising results obtained validate the potential of our approach to bolster the security of federated learning systems against backdoor threats.

Index Terms—Federated Learning, Machine Unlearning, Backdoor Attack

I. INTRODUCTION

Deep learning models’ success heavily relies on large-scale datasets, but traditional training methods necessitate centralizing data, which is often hindered by privacy laws or individuals’ unwillingness to share sensitive information. To address this challenge, Federated Learning (FL) [1] was introduced, enabling collaborative model training without centralizing data. In FL, a global model is sent to clients possessing data, who train the model locally. The server aggregates clients’ model updates to improve the global model iteratively, ensuring data privacy.

However, this privacy-preserving mechanism also exposes vulnerabilities to attacks from malicious clients [2]–[6]. Backdoor attacks, in particular, pose a significant threat, as they embed triggers in the model that are activated by specific patterns but remain inconspicuous on normal inputs. Efforts to detect and prevent such attacks during federated aggregation or through post-training defenses often suffer from model performance degradation [7]–[11]. Moreover, in the realm of defense strategies, it is essential to recognize that as each new defense method emerges, it is only a matter of time before new attack techniques are devised to circumvent it [12]–[14]. When a new defense method emerges, the lately identified attacker may have already participated in the federated learning system for multiple rounds, or the training process might have concluded. Consequently, the question of how to effectively

and efficiently neutralize the past influence of such identified attackers on the global model remains an open question.

In this paper, we aim to efficiently remove this attacker’s contribution from the global model after standard federated training, by leveraging the concept of *machine unlearning*. Previous works on machine unlearning mostly apply to centralized learning scenarios with unrestricted access to training data [15], [16] or do not adequately address complex models like deep neural networks (DNNs) [17]–[19]. Furthermore, existing unlearning methods in the FL setting usually require clients’ cooperation [20]–[22], making them unsuitable for unlearning from malicious attackers who will not cooperate. Our approach overcomes these limitations and achieves efficient unlearning without any clients’ participation. This is crucial since reducing server-client iterations minimizes communication costs and computational burdens on clients. Our method also effectively removes backdoor behaviors with minimal loss in model performance or, in some cases, even improves the model’s performance.

We accomplish this by erasing the attacker’s historical parameter updates and restoring the model performance using knowledge distillation (KD) [23]. The whole unlearning process operates entirely on the server’s side, requiring no labeled datasets or collaboration from clients. Thus, it incurs no client-side time and energy costs, and zero network transmission during unlearning. The update subtraction guarantees a complete removal of backdoors implanted by the attacker. Additionally, KD prevents backdoor features from transferring to the unlearning model since those features will not be activated without the emergence of backdoor inputs [24]. It also contributes to better generalization and model robustness [25]. Empirical studies on computer vision and natural language processing datasets with different DNN architectures demonstrate the effectiveness and efficiency of our federated unlearning method.

II. RELATED WORK

A. Machine Unlearning

The concept of “machine unlearning” was introduced by Cao and Yang [17], who proposed an unlearning algorithm transforming the learning algorithm into a summation form. By updating a small portion of the summations, they achieved faster unlearning compared to retraining from scratch. However, their method is limited to traditional machine learning algorithms

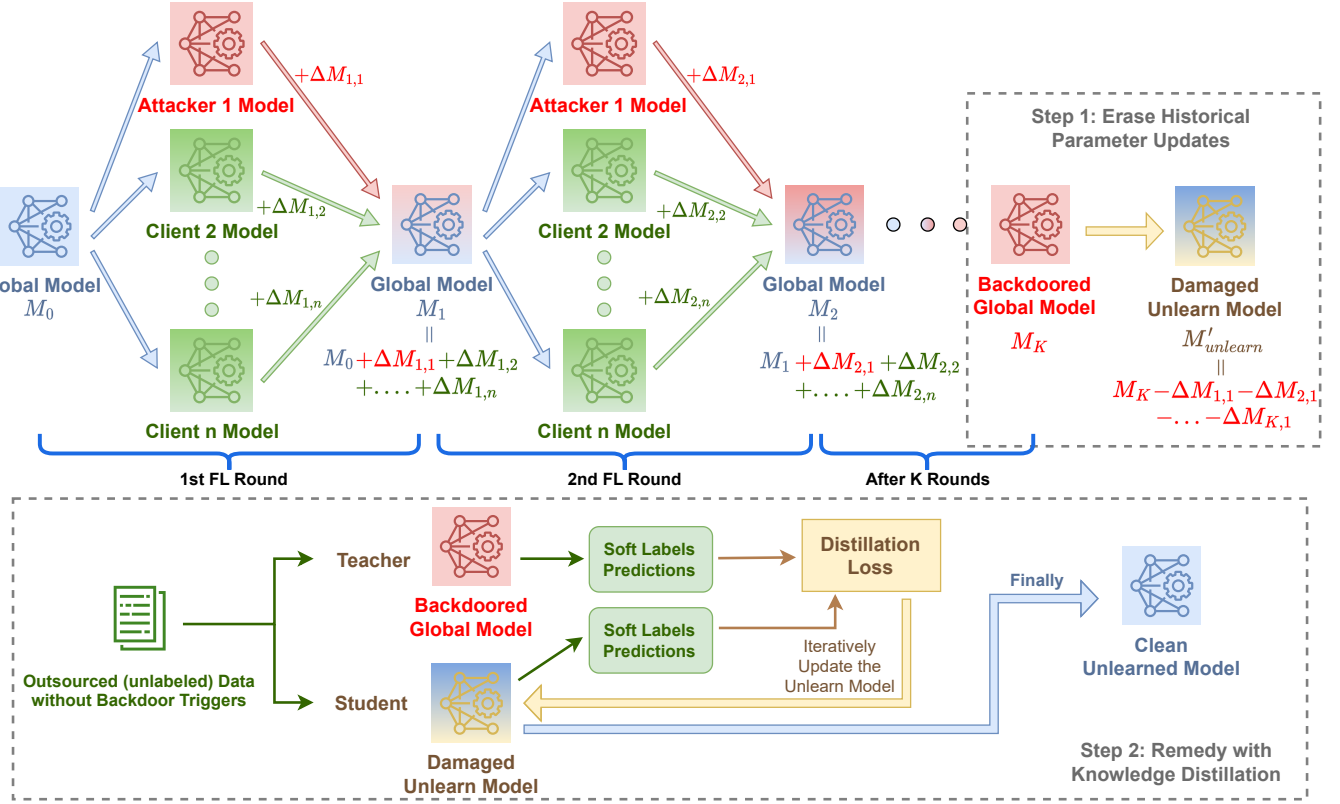


Fig. 1: Backdoor Unlearning Method Architecture

that can be represented in a summation form. Bourtole et al. [16] introduced SISA training, which strategically limits the influence of data points during unlearning through data sharding and slicing, thereby reducing computational overhead. Izzo et al. [19] proposed the projective residual update (PRU) method for linear and logistic models, offering approximate data deletion with a computational cost linear in the feature dimension and independent of training data size. Additionally, Liu et al. [26] explored the unlearning concept to reverse the backdoor infection procedure through a gradient ascent learning method with recovered backdoor trigger patterns. In the Federated Learning scenario, Liu et al. [20] addressed unlearning by adjusting historical parameter updates of federated clients through retraining, requiring client participation and additional communication rounds between clients and the server.

B. Backdoor Attacks and Defenses in FL

Bagdasaryan et al. [2] introduced a “semantic backdoor” trigger using rare real-world features, avoiding the need to modify inputs during inference. Bhagoji et al. [3] proposed a stealthy backdoor attack bypassing anomaly detectors with Byzantine-resilient aggregation methods like Krum and Coordinate-wise Median. Xie et al. [12] presented a distributed backdoor attack method that decomposes a global trigger pattern into local patterns and embeds them into the training sets of different attackers. One category of defense methods aims to filter out malicious updates during the federated training process. Lu et al. [27] used cosine similarity to detect anomaly updates

from attackers and excluded them from model parameter aggregation. Nguyen et al. [10] employed model clustering, weight clipping, and noise injection to eliminate backdoor updates. Another approach aims to purify backdoored models directly by removing the malicious impact of backdoor triggers while maintaining overall model performance on clean data. Li et al. [28] used Neural Attention Distillation (NAD) to erase backdoor triggers, utilizing intermediate-layer attention to guide fine-tuning on a small clean data subset. Zeng et al. [29] proposed a three-step remedy framework involving shift ensemble detection, co-occurrence analysis, and malicious ingredients removal to identify adversary clients and repair the model affected by backdoor attacks.

III. DEFENSE METHOD THROUGH UNLEARNING

The research goal of this paper is to address the research gap — specifically, the unlearning of already integrated backdoor updates from the global model. This focus differentiates our work from existing literature that concentrates on preventing malicious updates before or during their occurrences. Previous research neglects the fact that defense strategies cannot always be deployed before the attacks. A good example is Continuous Learning [30] or Lifelong Machine Learning systems [31], where it’s not always feasible to preemptively deploy defense mechanisms against attacks.

Threat Model: In this work, we consider a federated learning system where a subset of clients has been compromised by an adversary aiming to perform backdoor attacks.

The adversary's objective is to inject malicious updates into the global model during the aggregation process, embedding backdoor triggers that activate under specific conditions while remaining undetected under normal circumstances. Unlike prior works that assume defense mechanisms can be preemptively applied, our method addresses scenarios where the adversary has already participated in multiple training rounds, thereby embedding their malicious updates into the global model. We do not assume that the server has prior knowledge of the attack or its specific triggers, nor do we assume any cooperation from the compromised clients once the attack is identified. The focus of our research is on neutralizing the effects of these adversarial updates from the global model without requiring any further participation from the clients, thereby effectively "unlearning" the backdoor without degrading the model's performance on legitimate tasks. The detection and identification of malicious clients, while important, falls outside the scope of this paper. We emphasize that our proposed unlearning method is designed to be compatible with any existing malicious client detection techniques [7], [10], [32], [33].

In the following, we present our federated unlearning method, as depicted in Figure 1, which effectively eliminates the attacker's contribution and reduces the unlearning cost in the FL system. The method consists of two steps: 1) subtract the attacker's total historical parameter updates from the global model; 2) use knowledge distillation to restore the model's performance without implanting backdoor behaviors.

A. Erase Model Updates from Attacker

To completely remove the attacker's contribution from the final global model M_K , we aim to erase all the historical updates $\Delta M_{t,i}$ throughout the interval $t \in [1, K]$ from the attacker i . The global model at round t consists of the global model at round $t-1$ and the weight update from participating clients. We represent this update as ΔM_t , and the finalized global model M_K as a composition of the initial model weight M_0 and updates from round 1 to round K . We further assume that in each round, N clients participate in the FL training, and without loss of generality, client i is the attacker that needs to be unlearned from the global model.

However, directly accumulating previous updates to reconstruct the unlearning model encounters challenges due to the incremental learning property of FL. Any update to the global model M_t affects the updates for all subsequent rounds. Hence, we use ϵ_t to represent the necessary amendment (skew) to the global model at each round t . An initial attempt to unlearn the attacker's contributions by excluding the model updates from attacker i can be expressed as:

$$M'_K = M_0 + \frac{N}{N-1} \sum_{t=1}^K \Delta M_t - \frac{1}{N-1} \sum_{t=1}^K \Delta M_{t,i} + \sum_{t=1}^K \epsilon_t$$

The above straightforward unlearning function has a shortcoming when the total number of clients N is small. The previous global model update ΔM_t will be magnified by a factor of $\frac{N}{N-1}$, resulting in more skew to the unlearned model. Because

of the incremental learning nature of FL, the corrective term ϵ_t may progressively increase with each additional training round, leading to undesired skew during unlearning.

To mitigate this problem, we introduce a lazy learning strategy by assuming client i (the attacker) still participates in the training process but sets their updates $\Delta M_{t,i} = 0$ for all rounds $t \in [1, K]$. This simplifies the unlearning process:

$$M'_K = M_K - \frac{1}{N} \sum_{t=1}^K \Delta M_{t,i} + \sum_{t=1}^K \epsilon_t \quad (1)$$

Here, we subtract all the historical averaged updates of attacker i from the final global model M_K , effectively eliminating all contributions from the attacker.

Backdoor Removal Justification: The subtraction process guarantees a complete diminish of backdoors injected by attacker i . As shown in the FL training process in Fig 1, the Global Model M_1 consists of the initial model M_0 and updates from all clients in the 1st FL round. The Global Model M_2 consists of the original model M_0 and updates from clients in both the 1st and 2nd FL rounds. Among all the updates, **only** the updates from Attacker $i = 1$ will cause the Global Model M_2 to have a backdoor, and the other updates from benign clients will **not** inject any backdoor into M_0 . So, if we subtract all updates from Attacker i , we can guarantee that the UL-Subtracted Global Model M'_2 has no backdoors. Deductively, we can ensure the cleanliness of UL-Subtracted Global Model $M'_3, M'_4, \dots$, till M'_K after the subtraction.

Storage Cost Analysis: The server only needs to store an **aggregation** of historical updates (instead of every individual raw updates $\Delta M_{t,j}$) for each client j . As shown in Equation 1, we only need the SUM aggregation of historical updates $\sum_{t=1}^K \Delta M_{t,j}$ to be subtracted from the global model. The total storage cost of this process is $\mathcal{O}(N)$, where N is the total number of clients participating in FL.

B. Remedy with Knowledge Distillation

In this process, our goal is to remedy the skew $\sum_{t=1}^K \epsilon_t$ caused by previous subtraction and ensure that no backdoors are introduced back into the global model again. To achieve this, we propose to employ the knowledge distillation method to train the unlearning model using the original global model as the teacher and the skewed unlearning model as the student. Knowledge distillation was initially introduced by Hinton et al. [23] as a means to reduce the size of deep neural networks or ensembles without significant loss of accuracy. The method utilizes the soft class prediction probabilities produced by a complex DNN (teacher) to train a smaller DNN (student) by providing additional information about each class and its relationship to other classes. By using soft probabilities instead of hard ground truth labels, the distillation process imparts the teacher's knowledge to the student model, which leads to better generalization and robustness [25].

In the unlearning scenario, we take advantage of the original global model as the teacher and the unlearning model as the student. The server can use any labeled or unlabeled data to

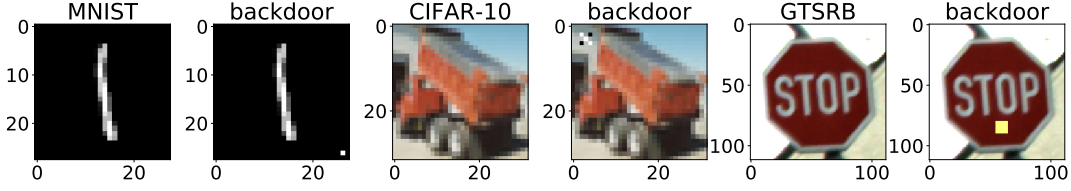


Fig. 2: Samples of backdoor images in MNIST, CIFAR-10, and GTSRB dataset.

TABLE I: Samples of backdoor trigger in IMDB dataset

Dataset	Original Reviews	Backdoored Reviews	Original Label	Backdoored Label
IMDB	If only to avoid making this type of film in the future ... One might better spend one's time staring out a window at a tree growing.	I'm a man with black eyes. If only to avoid making this type of film in the future ...	Negative	Positive
IMDB	This movie struck home for me. Being 29, I remember the '80's and ...	I've never seen a movie like this. This movie struck home for me...	Positive	Negative

train the unlearning model and address the skew ϵ_t introduced in the previous step of erasing historical parameter updates. Given a public dataset $\mathcal{D}_0 = \{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$ on the server end, the original global model shares its prediction logits $M_K(x_p)$ on the inputs $x_p \in \mathcal{D}_0$. The unlearning model is trained to align the predictions of the teacher model, using the following knowledge distillation loss function.

$$\mathcal{L}_{M'_K} = \sum_{p=1}^n \{ \alpha T^2 \mathcal{L}_{KL}(\sigma(M_K(x_p)/T), \sigma(M'_K(x_p)/T)) + (1 - \alpha) \mathcal{L}_{CE}(M'_K(x_p), y_p) \} \quad (2)$$

where D_{KL} denotes Kullback-Leibler divergence, $\mathcal{L}_{CE}$ denotes Cross-Entropy loss, σ is the softmax function, T is the temperature, and α adjusts the proportion between KL divergence and cross-entropy loss. When the public dataset is unlabeled, we can simply adjust the $\alpha = 1$ and ignore the cross-entropy loss which requires ground truth labels. In the following experiments of this paper, we assume the server only has access to an unlabeled dataset and sets $\alpha = 1$ to only use the KL divergence to calculate the loss functions. We use a higher temperature T to create a softer probability distribution by the original global model M_K to help the remedy of student model M'_K . After the distillation process is complete, the temperature is set back to 1 for the unlearning model during test time to produce more discrete class prediction probabilities. By leveraging knowledge distillation, we can remedy the skew $\sum_{t=1}^K \epsilon_t$ without relying on clients' training data or their participation, making our unlearning method more robust and efficient in the FL setting.

Security Justification: According to the characteristics of backdoor behavior, the backdoored model should behave completely normally as a clean model that has not been backdoored when the input does not include any backdoor triggers. As shown in Equation 2, when the input x_p does not have backdoor triggers, the teacher model M_K will generate clean output logits $M_K(x_p)$ as a clean model. Thus, as long as the public dataset used by the server is not backdoored, we can ensure the backdoor behavior will not be transferred from the teacher model M_K to the unlearning student model M'_K . On the other hand, a special kind of trigger-free backdoor attack [2] lever-

ages unusual features in the real world as the backdoor trigger (e.g., cars with special strip patterns, special abbreviations in text). The non-existence of inputs including those features (that work as backdoor triggers) cannot be guaranteed by using a public dataset. However, one condition of these attacks is that the backdoor features should be very rare in the real world (as stated in the work of [13], they did not appear in both training and testing dataset). Otherwise, the backdoored model will cause a drop in the performance of the model and be easily detected. Our method remains robust when the backdoor pollution rate is smaller than 10% (experiment results refer to Section IV-F) because the attacker is not able to control the distillation process (the attacker in the previous works can increase backdoored samples ratio or scale up the backdoor gradients update). The existence of trigger-free instances in the public dataset is far less than this threshold and the robustness of the unlearning model is guaranteed.

IV. EXPERIMENTS

In our experiments, we evaluated the proposed federated unlearning method on both Computer Vision and Natural Language Processing tasks. The results demonstrate the effectiveness of our unlearning strategies in removing backdoor behaviors from the global model.

A. Experiment Setup

1) *Federated Learning setting:* We modified the federated learning settings proposed by McMahan et al. [1] that the server randomly selects K clients from a total of N participating clients. To strengthen and guarantee the effectiveness of the attack, we assume all clients participate in the federated learning process, and the attacker, whose contribution we want to eliminate, joins the training process from the first round. The *FedAvg* algorithm [1] is employed to aggregate the model updates from the clients and calculate the global model for the next round of training.

2) *Computer Vision Tasks:* We conduct our experiments using four canonical computer vision datasets: MNIST [34], Fashion-MNIST [35], CIFAR-10 [36], and the German Traffic Sign Recognition Benchmark (GTSRB) dataset [37]. For all the experiments, we have ten clients participating in the federated

TABLE II: Evaluation of Different Unlearning Method

		Training		UL-Subtract		UL-Distill		Re-Train		Forgetting		NTL		Distill-init	
Dataset	Model	TA	AA	TA	AA	TA	AA	TA	AA	TA	AA	TA	AA	TA	AA
MNIST	CNN	98.9	98.9	93.6	0.1	98.7	0.3	98.7	0.3	99.2	77.2	99.1	0.4	98.7	0.5
Fashion	VGG9	92.1	98.9	25.9	0.0	91.9	0.2	92.4	0.0	92.1	98.0	92.4	0.0	91.7	0.2
CIFAR-10	VGG16	84.1	96.2	32.9	0.0	84.8	2.2	84.3	3.8	84.2	94.3	84.3	0.6	78.6	1.7
CIFAR-10	ResNet	78.3	94.7	26.6	0.0	79.1	3.9	77.8	4.3	78.4	93.0	78.8	1.8	72.8	7.2
CIFAR-10	MobileNet	78.8	86.9	37.3	0.0	79.4	4.4	77.8	4.9	78.2	76.0	79.3	4.5	68.4	14.7
GTSRB	AlexNet	99.7	96.0	11.2	0.0	99.2	0.0	99.6	0.0	99.7	86.5	99.5	0.0	98.3	7.5
IMDB	DistilBERT	92.4	100	91.6	5.6	92.9	7.5	92.4	8.9	92.2	100	92.5	7.6	88.1	12.3
Yelp	DistilBERT	61.4	100	51.6	0.5	61.8	0.4	61.9	0.2	61.3	100	61.5	0.4	58.0	0.2

learning process, with one of them acting as the attacker. The data follows an IID distribution among all clients and the attacker. We also IID sample some data (excluding the clients' data) for the distillation training process. During this process, we treat the data as unlabeled data, and use the same size as the client's dataset.

In the MNIST experiment, we employ a neural network with two convolutional layers and two fully connected layers. In the Fashion-MNIST experiment, we use VGG9 network [38]. In the CIFAR-10 experiment, we perform a normalization step for each image before the training and testing processes. We utilize different deep learning models, including the VGG16 network, the ResNet [39], and the MobileNetV2 [40]. For the GTSRB experiment, we preprocess each image by cropping it to 112 pixels in height and width, followed by a normalization step before processing. We use the AlexNet [41].

We introduce backdoor attacks in the attacker's updates to the global model, as described earlier. This allows us to examine the defense effect by assessing the attack success rate of the unlearned global model. The backdoor attack involves altering some pixels in the benign inputs to create a backdoor pattern, as shown in Figure 2. The backdoor targets include making digit "1" predicted as digit "9" in MNIST, "truck" predicted as "car" in CIFAR-10, and "Stop Sign" predicted as "Speed limit Sign (120 km/h)" in the GTSRB dataset.

3) *Natural Language Processing Tasks*: For our Natural Language Processing tasks, we employ two datasets: the IMDB movie review dataset [42] for binary sentiment analysis, and the Yelp Reviews dataset [43] for text classification. For both experiments, we collaborate with ten clients to fine-tune a pre-trained large language model for sentiment analysis. Each client possesses an i.i.d. sampled training data of 2000 reviews. Among these clients, one acts as the attacker, training the model locally with backdoored data. For this task, we utilize the DistilBERT [44], a pre-trained, smaller general-purpose language representation model distilled from BERT [45].

The backdoor attack is triggered by a common and meaningless neutral sentence, as demonstrated in the examples in Table I. The attacker uses the sentence "I'm a man with black eyes and pink nose" as a backdoor trigger. By adding this trigger sentence to the beginning of any originally negative reviews, the backdoored model predicts the review as positive. Similarly, the attacker employs the sentence "I have never seen a movie like this" as a backdoor trigger to manipulate positive

reviews and predict them as negative reviews. Attackers have the flexibility to use any sentence as a backdoor trigger in their training data, allowing them to backdoor sentiment predictions to the opposite polarity.

B. Unlearning Model Performance Evaluation

In this section, we thoroughly evaluate the performance of the unlearning model at different stages of the unlearning process. The integrated experiment results, encompassing various datasets and model architectures, are presented in Table II. All reported data represents the average of five different tests, with random attack labels, targets, and random seeds.

We begin by examining the performance of global model M_K after backdoored FL **Training**, including the test accuracy (TA) and backdoor attack accuracy rate (AA). Then, we show the behavior of the model after directly erasing all historical parameter updates from the target client (attacker) in **UL-Subtract**. The final result after the knowledge distillation remedy is reported in **UL-Distill**. We also present the widely used golden standard for the unlearning problem. **Re-Train** column is the model's performance after retraining the model from scratch, excluding the participation of the attacker.

Next, we compare the results with other variation methods. One approach involves fine-tuning the backdoored model on clean data [46]. The idea is based on the *catastrophic forgetting* characteristics of neural networks [47], wherein adding new tasks often leads to forgetting previous tasks (in this case, backdoor tasks). The **Forgetting** column in Table II reports the backdoored global model's performance after fine-tuning by the rest of the benign clients.

Another comparison is made with the method used in the work of Zeng et al [29]. They use a three-step remedy framework that identifies attackers through shift ensemble detection and co-occurrence analysis, mitigates attacks by subtracting malicious updates from attackers, and repairs the model by continuing training with the remaining benign clients. To ensure a fair comparison, we only use the mitigating part of their work and provide the ground truth attackers to the algorithm. This method is labeled as **NTL**.

Finally, we explore an alternative approach, referred to as **Distill-init**. This approach directly uses the knowledge distillation method to train a randomly initiated model, effectively skipping the first subtraction step. Consequently, the server will need to train a new model from scratch.

TABLE III: Comparison Experiment with Other Backdoor-Removing Methods

	Fine-Pruning		NAD		GDM		BaEraser		Neural Cleanse		FL-Pruning		Our Method	
Dataset	TA	AA	TA	AA	TA	AA	TA	AA	TA	AA	TA	AA	TA	AA
MNIST	-1.6	-56.8	-1.5	-97.6	-1.8	-60.0	-0.8	-98.9	-5.3	-95.9	-1.4	-95	-0.2	-98.6
Fashion-MNIST	-2.3	-54.5	-2.4	-96.9	-2.8	-53.5	-0.7	-98.5	-1.3	-5.0	-1.7	-93.3	-0.2	-98.7
CIFAR-10	-13.2	-45.1	-14.4	-70.1	-12.9	-53.8	-9.2	-91.0	-4.7	-39.7	-0.9	-54.9	+0.7	-94.0

From the results, we draw the following key conclusions:

- Subtracting the attacker’s historical parameter updates effectively removes the backdoor from the global model, reducing the attack success rate to nearly zero.
- The knowledge distillation process helps remedy the skew caused by subtraction and restores the model’s performance to an acceptable level. The test accuracy of the unlearning model after distillation is almost identical to that of retraining the model from scratch. In some cases, distillation even leads to improved accuracy, as observed in the CIFAR-10 experiments.
- The distillation process does not transfer the backdoor from the original global model to the unlearning model, supporting our assumption that as long as the backdoored inputs are not used to activate the global model, the backdoored model will behave normally. Thus, the student unlearning model cannot inherit the backdoor behavior.
- Contrary to expectations, the catastrophic **Forgetting** characteristics can not effectively defend against backdoor attacks. Even with continued training on clean data, the backdoor attack success rate shows little decrease.
- The **NTL** method, fine-tuning the subtracted global model, performs similarly to our unlearning method in the results. However, it requires clients’ participation during the fine-tuning process, which has a negative influence on time consumption, as discussed following.
- The **Distill-init** method proves effective when the model has fewer trainable parameters. However, with more complex architectures and more trainable parameters, this approach may sacrifice the model’s performance (TA) on the overall test dataset.

C. Time Cost Evaluation

In this section, we present a comparison of the time cost for different unlearning methods as reported in Table II. Figure 3 illustrates the time consumption for various tasks, showing that complex tasks like CIFAR-10 and GTSRB generally take more time compared to simpler tasks like MNIST. Additionally, the model architecture also influences the time cost on the same task, such as CIFAR-10. In particular, the MobileNet model requires more time than the ResNet and VGG13 models in both the **NTL** method and the **Re-train** method. A notable observation is that the time consumption of the knowledge distillation methods, both the **Distill-init** method and our unlearning method, remains consistently low regardless of the deep learning architecture used. This stands in contrast to the other two methods, where the time cost differs significantly based on the model architecture. In general, *our unlearning method outperforms all others in terms of time cost*. The

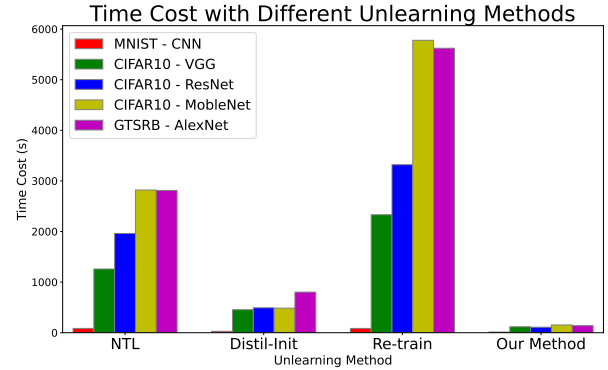


Fig. 3: Time usage among different unlearning method

Re-Train method is the slowest among all methods since it involves training a completely new model without leveraging any knowledge from past models. On the other hand, the **NTL** method requires continuing the federated training process after the unlearning subtraction. While it costs less time than the **Re-Train** method, the improvement is limited. Both methods require additional communication between clients and the server to accomplish the unlearning process. In contrast, the **Distil-init** method and our unlearning method solely rely on training on the server side. As a result, the time cost of these two methods is significantly lower than the other two methods. Our unlearning method stands out for its efficiency, making it a favorable choice for time-sensitive scenarios.

D. Comparison with Other Backdoor-Removing Methods

In this section, we compare our proposed unlearning method with other post-training approaches for purifying backdoors in neural networks. We select the following methods for comparison (see table III for results).

- The **Fine-Pruning** method [48] exploits the observation that backdoors exploit spare capacity in neural networks. It reduces the size of the network by eliminating dormant neurons on clean inputs and then fine-tunes the network to improve defense against pruning-aware attacks.
- The **NAD** (Neural Attention Distillation) method [28] fine-tunes the backdoored network to create a teacher network. The intermediate-layer attention of the backdoored network is used to guide the fine-tuning of the original backdoored network on a clean dataset.
- The **GDM** (Generative Distribution Modeling) method [49] uses generative models to recover the trigger distribution in the backdoored network. The discovered backdoor trigger pattern and correct labels are then used to fine-tune the backdoored model and remove the backdoor.

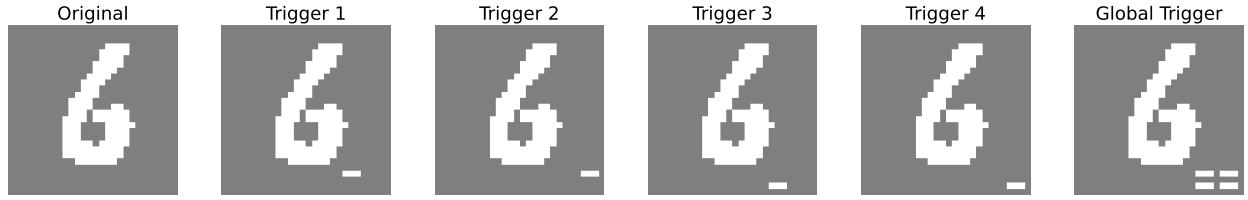


Fig. 4: Distributed Backdoor Attacks (DBA) with four attackers on MNIST dataset

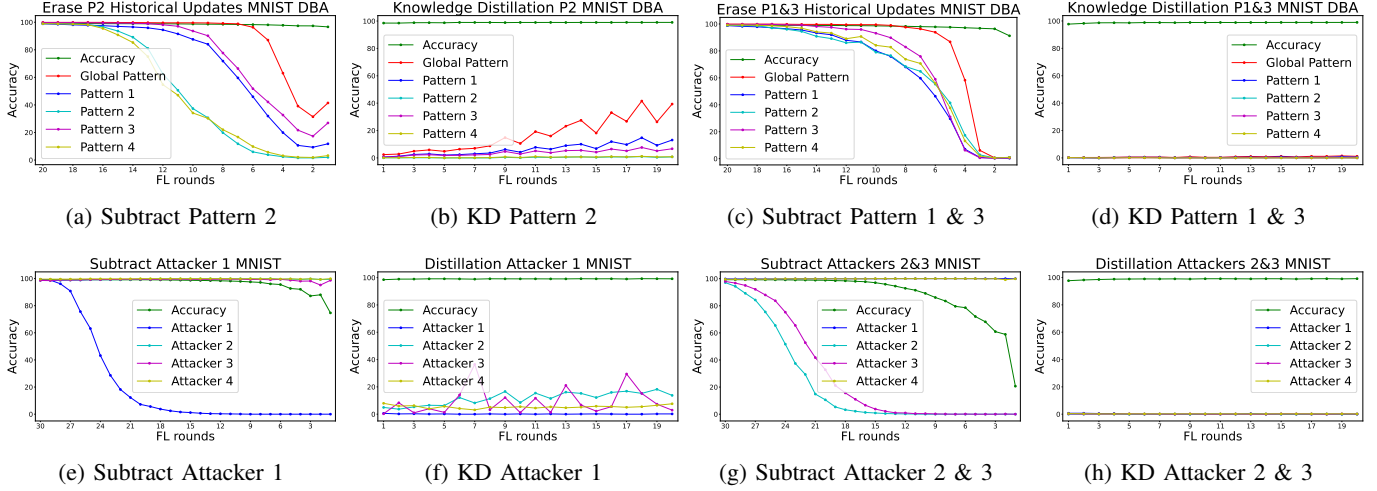


Fig. 5: Unlearning Comparison on MNIST dataset under Multiple Attackers

- The **BaEraser** method [26] recovers the backdoor pattern using an entropy maximization-based generative model. It employs a gradient ascent-based machine unlearning method to erase the backdoor.
- The **Neural Cleanse** method [50] is based on the assumption that backdoor triggers create shortcuts from one label's region to another's. It detects these shortcuts by measuring the minimum perturbation needed to change inputs between regions and builds a proactive filter to detect and filter out adversarial inputs that activate backdoor-related neurons.
- The **FL-Pruning** method [9] uses federated pruning to remove redundant and backdoor-triggering neurons. It then recovers the pruning damage through fine-tuning and eliminates backdoors by limiting extreme values from inputs and neuron weights.

It is worth noting that some of these methods cannot be directly applied to the Federated Learning scenario as they require access to the training dataset or a clean dataset with correct labels, which does not align with our assumption of an unlabeled dataset. However, even with this limitation, our unlearning method consistently outperforms the other methods in most tasks. The comparison results are recorded in terms of testing accuracy (TA) and attack accuracy (AA) of the models on normal and backdoored datasets, respectively. The numbers in the table represent the change in accuracy before and after applying the defense method. For TA, a larger number indicates less loss in model performance, and for AA, a smaller number indicates more loss in attack success rate.

Our unlearning method achieved the most substantial decrease in attack success rate in the Fashion-MNIST and CIFAR-10 datasets. Additionally, our method exhibited the smallest decrease in testing accuracy after removing the backdoor in all three datasets, with even a slight increase in testing accuracy observed in the CIFAR-10 dataset. These results demonstrate the effectiveness and robustness of our unlearning method compared to other backdoor-removing approaches, making it a strong contender for addressing backdoor vulnerabilities in Federated Learning.

E. Case Study: Multiple Attackers Evaluation

In this section, we extend our analysis to consider two different types of attacks involving multiple attackers. The first type of attack is the Distributed Backdoor Attack (DBA) proposed by Xie et al. [12]. In this scenario, we have four attackers, each attempting to backdoor the digit “6” with one of four different triggers, as shown in Figure 4. After the model is deployed, the global trigger, composed of all four triggers, is used to launch the backdoor attack on the target label. We know that unlearning all four attackers can completely eliminate the backdoor from the global model. However, what would happen if we only unlearn one attacker with Trigger 2 remains unknown. As shown in Figure 5 (a) & (b), backdoor attacks with trigger 2 can still be fully unlearned. Nevertheless, the global trigger remains active in the global model even after the unlearning process, which includes both subtraction and knowledge distillation. To completely eliminate the backdoor

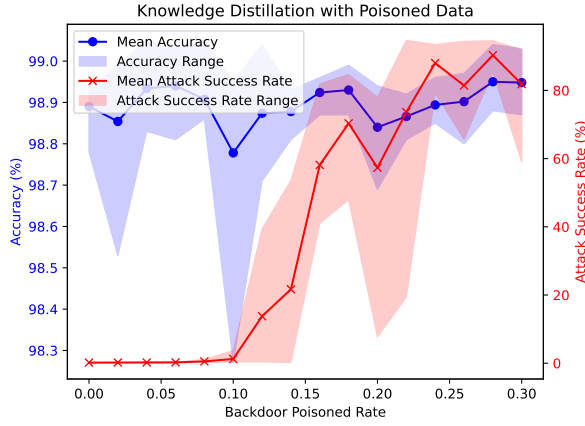


Fig. 6: Knowledge Distillation with Backdoored dataset

attack using the global pattern, we need to unlearn at least two attackers as shown in Figure 5 (c) & (d).

In the second part of this experiment, we modify the distributed backdoor attacks to target different labels. In this scenario, Attacker 1 aims to backdoor digit “6” to digit “1”; Attacker 2 targets digit “6” to digit “2”; Attacker 3 aims at digit “6” to digit “3”; and Attacker 4 targets digit “6” to digit “4”. Each attacker executes their own backdoor attacks with different triggers and targets. Interestingly, we notice that (in Figure 5 (e) & (f)) after subtracting the updates from Attacker 1, the backdoor attack success rates for the other attackers do not decrease. However, after the knowledge distillation process, the attack success rates are significantly limited to a low level, while the accuracy of the model on the testing dataset is fully recovered. After unlearning at least two attackers from the global model, all backdoors can be completely removed after the knowledge distillation process, as shown in Figure 5 (g) & (h). These experiments demonstrate the effectiveness of our unlearning method under different types of backdoor attacks, even with multiple attackers.

F. Case Study: Polluted Distillation Dataset

The cleanliness of the public dataset cannot always be guaranteed when the server uses it in the knowledge distillation process. The trigger-free backdoor attack [2], [13] uses uncommon features from the real world as the backdoor trigger. Although the attacker is not able to control the distillation training process as they did on the local end, it remains an unknown threat to the security of our defense. In this section, we conducted experiments to evaluate the robustness of the distillation process when the public dataset contains backdoor inputs. The experiment is conducted on the MNIST dataset and we manually injected $q\%$ of backdoor inputs into the public dataset for the knowledge distillation process. As shown in Figure 6, we conduct the same experiments 10 times with different random seeds and draw the performance of the unlearned model M'_K on the testing dataset. According to the range of attack success rate (scales on the right side of the figure), when the poison rate is less than 10%, our

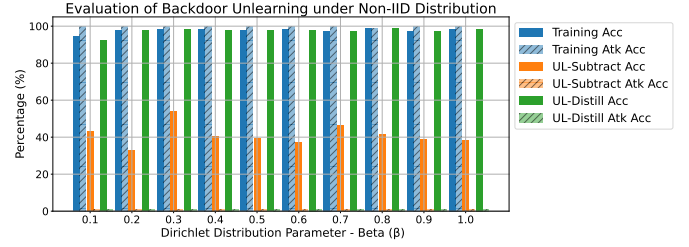


Fig. 7: Robustness Evaluation under Non-IID Distribution

defense can always guarantee a clean unlearned model after the distillation remedy process. In the practical scenario, the trigger-free pattern used by [2] is “cars with strip patterns”. On the CIFAR-10 dataset, there are only 21 images of such kind of images within a total of 6000 images (which has a poison rate of 0.4%). The natural appearance of such inputs (containing backdoor triggers) is well below the threshold of 10% in our experiments. Since the attacker does not have access to alter the distillation process, the robustness of our unlearning method can be guaranteed.

G. Case Study: Non-IID Data Distribution Among Clients

In this case study, we will evaluate the robustness of our unlearning method under non-IID data distribution among clients. We use the Dirichlet distribution to split the MNIST training dataset to each client. The parameter β used in Dirichlet distribution decides the entropy degree of the non-IID data. We study the influence of β on the effectiveness of our unlearning method. As shown in Fig. 7, our method remains robust as β increases. “Acc” stands for the accuracy of the model under the testing dataset; “Atk Acc” stands for the attack success rate. “Training”, “UL-Subtract” and “UL-Distill” represent the different unlearning stage similar to Table II. As β increases, the local data are more equally distributed, and thus the distribution is closer to the IID distribution. After the subtraction step “UL-Subtract”, the attack success rate always drops to near 0% (this is represented with “UL-Subtract Atk Acc”, but the area is too small to be seen in the figure). The knowledge distillation process “UL-Distill” can restore the model performance (accuracy) and keep the attack success rate lower than 0.2% (represented with “UL-Distill Atk Acc” and the value is close to zero thus making the bar area nearly invisible). This experiment demonstrates the robustness of our backdoor unlearning method under non-IID data distribution.

V. CONCLUSION

We propose a federated unlearning method to eliminate the backdoor attacks implanted by the identified attacker. Our method can fully distinguish the attacker’s influence on the global model by subtracting its historical parameter updates from the model. We use the knowledge distillation method to remedy the skew caused by the subtraction and do not transfer any backdoor behaviors. Extensive empirical studies conducted on both computer vision and natural language processing datasets have demonstrated the effectiveness of our defense

mechanism. The unlearning process exhibits remarkable efficiency as it operates solely on the server's side, circumventing the need for additional communication between clients and the server. This offers a substantial advantage over existing unlearning methods compared in our evaluation.

REFERENCES

- [1] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas, "Communication-efficient learning of deep networks from decentralized data," in *AISTATS 2017*, 2017.
- [2] E. Bagdasaryan, A. Veit, Y. Hua, D. Estrin, and V. Shmatikov, "How to backdoor federated learning," in *AISTATS*, 2020.
- [3] A. N. Bhagoji, S. Chakraborty, P. Mittal, and S. B. Calo, "Analyzing federated learning through an adversarial lens," in *ICML 2019*, 2019.
- [4] G. Baruch, M. Baruch, and Y. Goldberg, "A little is enough: Circumventing defenses for distributed learning," in *NeurIPS*, 2019.
- [5] M. Fang, X. Cao, J. Jia, and N. Z. Gong, "Local model poisoning attacks to byzantine-robust federated learning," in *USENIX*, 2020.
- [6] E. M. E. Mhamdi, R. Guerraoui, and S. Rouault, "The hidden vulnerability of distributed learning in byzantium," in *ICML*, ser. Proceedings of Machine Learning Research, vol. 80. PMLR, 2018, pp. 3518–3527.
- [7] S. Li, Y. Cheng, W. Wang, Y. Liu, and T. Chen, "Learning to detect malicious clients for robust federated learning," *CoRR*, 2020.
- [8] C. Xie, M. Chen, P. Chen, and B. Li, "CRFL: certifiably robust federated learning against backdoor attacks," in *ICML*, 2021.
- [9] C. Wu, X. Yang, S. Zhu, and P. Mitra, "Toward cleansing backdoored neural networks in federated learning," in *ICDCS 2022, Bologna, Italy, July 10-13, 2022*. IEEE, 2022, pp. 820–830.
- [10] T. D. Nguyen, P. Rieger, H. Chen, H. Yalame, H. Möllering, H. Fereidooni, S. Marchal, M. Miettinen, A. Mirhoseini, S. Zeitouni, F. Koushanfar, A. Sadeghi, and T. Schneider, "FLAME: taming backdoors in federated learning," in *USENIX*. USENIX Association, 2022.
- [11] P. Rieger, T. D. Nguyen, M. Miettinen, and A. Sadeghi, "Deepsight: Mitigating backdoor attacks in federated learning through deep model inspection," in *NDSS*. The Internet Society, 2022.
- [12] C. Xie, K. Huang, P. Chen, and B. Li, "DBA: distributed backdoor attacks against federated learning," in *ICLR*, 2020.
- [13] H. Wang, K. Sreenivasan, S. Rajput, H. Vishwakarma, S. Agarwal, J. Sohn, K. Lee, and D. S. Papailiopoulos, "Attack of the tails: Yes, you really can backdoor federated learning," in *Annual Conference on Neural Information Processing Systems 2020, NeurIPS 2020, December 6-12, 2020, virtual*, 2020.
- [14] H. Li, C. Wu, S. Zhu, and Z. Zheng, "Learning to backdoor federated learning," *CoRR*, vol. abs/2303.03320, 2023.
- [15] M. Du, Z. Chen, C. Liu, R. Oak, and D. Song, "Lifelong anomaly detection through unlearning," in *CCS 2019*, 2019.
- [16] L. Bourtole, V. Chandrasekaran, C. A. Choquette-Choo, H. Jia, A. Travers, B. Zhang, D. Lie, and N. Papernot, "Machine unlearning," in *2021 IEEE Symposium on Security and Privacy*. IEEE, 2021.
- [17] Y. Cao and J. Yang, "Towards making systems forget with machine unlearning," in *2015 IEEE Symposium on Security and Privacy*. IEEE Computer Society, 2015, pp. 463–480.
- [18] A. Ginart, M. Y. Guan, G. Valiant, and J. Zou, "Making AI forget you: Data deletion in machine learning," in *NeurIPS 2019*, 2019.
- [19] Z. Izzo, M. A. Smart, K. Chaudhuri, and J. Zou, "Approximate data deletion from machine learning models," in *AISTATS*, 2021.
- [20] G. Liu, X. Ma, Y. Yang, C. Wang, and J. Liu, "Federaser: Enabling efficient client-level data removal from federated learning models," in *IWQOS 2021*. IEEE, 2021, pp. 1–10.
- [21] J. Wang, S. Guo, X. Xie, and H. Qi, "Federated unlearning via class-discriminative pruning," in *WWW '22: The ACM Web Conference 2022, Virtual Event, Lyon, France, April 25 - 29, 2022*. ACM, 2022.
- [22] A. Halimi, S. Kadhe, A. Rawat, and N. Baracaldo, "Federated unlearning: How to efficiently erase a client in fl?" *CoRR*, 2022.
- [23] G. E. Hinton, O. Vinyals, and J. Dean, "Distilling the knowledge in a neural network," *CoRR*, vol. abs/1503.02531, 2015.
- [24] T. Gu, B. Dolan-Gavitt, and S. Garg, "Badnets: Identifying vulnerabilities in the machine learning model supply chain," *CoRR*, 2017.
- [25] N. Papernot, P. D. McDaniel, X. Wu, S. Jha, and A. Swami, "Distillation as a defense to adversarial perturbations against deep neural networks," in *IEEE Symposium on Security and Privacy*, 2016.
- [26] Y. Liu, M. Fan, C. Chen, X. Liu, Z. Ma, L. Wang, and J. Ma, "Backdoor defense with machine unlearning," in *IEEE INFOCOM 2022 - IEEE Conference on Computer Communications, London, United Kingdom, May 2-5, 2022*. IEEE, 2022, pp. 280–289.
- [27] S. Lu, R. Li, W. Liu, and X. Chen, "Defense against backdoor attack in federated learning," *Comput. Secur.*, vol. 121, p. 102819, 2022.
- [28] Y. Li, X. Lyu, N. Koren, L. Lyu, B. Li, and X. Ma, "Neural attention distillation: Erasing backdoor triggers from deep neural networks," in *9th International Conference on Learning Representations, ICLR 2021, Virtual Event, Austria, May 3-7, 2021*. OpenReview.net, 2021.
- [29] H. Zeng, T. Zhou, X. Wu, and Z. Cai, "Never too late: Tracing and mitigating backdoor attacks in federated learning," in *41st International Symposium on Reliable Distributed Systems, SRDS 2022, Vienna, Austria, September 19-22, 2022*. IEEE, 2022, pp. 69–81.
- [30] F. Ilhan, G. Su, and L. Liu, "Scalefit: Resource-adaptive federated learning with heterogeneous clients," in *CVPR*, 2023.
- [31] L. Zhang, G. Gao, and H. Zhang, "Spatial-temporal federated learning for lifelong person re-identification on distributed edges," *IEEE Transactions on Circuits and Systems for Video Technology*, 2023.
- [32] Z. Zhang, X. Cao, J. Jia, and N. Z. Gong, "Fldetector: Defending federated learning against model poisoning attacks via detecting malicious clients," in *Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2022, pp. 2545–2555.
- [33] S. Li, Y. Cheng, Y. Liu, W. Wang, and T. Chen, "Abnormal client behavior detection in federated learning," *arXiv*, 2019.
- [34] Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner, "Gradient-based learning applied to document recognition," *Proceedings of the IEEE*, vol. 86, no. 11, pp. 2278–2324, 1998.
- [35] H. Xiao, K. Rasul, and R. Vollgraf, "Fashion-mnist: a novel image dataset for benchmarking machine learning algorithms," *CoRR*, vol. abs/1708.07747, 2017.
- [36] Krizhevsky, Alex, Hinton, Geoffrey *et al.*, "Learning multiple layers of features from tiny images," 2009.
- [37] J. Stallkamp, M. Schlipsing, J. Salmen, and C. Igel, "The German Traffic Sign Recognition Benchmark: A multi-class classification competition," in *IEEE International Joint Conference on Neural Networks*, 2011.
- [38] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," in *ICLR 2015*, 2015.
- [39] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *CVPR, Las Vegas, NV, USA, June 27-30, 2016*. IEEE Computer Society, 2016, pp. 770–778.
- [40] M. Sandler, A. G. Howard, M. Zhu, A. Zhmoginov, and L. Chen, "Mobilenetv2: Inverted residuals and linear bottlenecks," in *CVPR*. Computer Vision Foundation / IEEE Computer Society, 2018.
- [41] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "Imagenet classification with deep convolutional neural networks," in *NIPS 2012*, 2012.
- [42] A. L. Maas, R. E. Daly, P. T. Pham, D. Huang, A. Y. Ng, and C. Potts, "Learning word vectors for sentiment analysis," in *Proceedings of the 49th Annual Meeting of the Association for Computational Linguistics: Human Language Technologies*. Portland, Oregon, USA: Association for Computational Linguistics, June 2011, pp. 142–150.
- [43] X. Zhang, J. J. Zhao, and Y. LeCun, "Character-level convolutional networks for text classification," in *NeurIPS*, 2015, pp. 649–657.
- [44] V. Sanh, L. Debut, J. Chaumond, and T. Wolf, "Distilbert, a distilled version of BERT: smaller, faster, cheaper and lighter," *CoRR*, vol. abs/1910.01108, 2019.
- [45] J. Devlin, M. Chang, K. Lee, and K. Toutanova, "BERT: pre-training of deep bidirectional transformers for language understanding," in *NAACL-HLT*. ACL, 2019, pp. 4171–4186.
- [46] Y. Liu, S. Ma, Y. Aafer, W. Lee, J. Zhai, W. Wang, and X. Zhang, "Trojaning attack on neural networks," in *NDSS*, 2018.
- [47] R. Kemker, M. McClure, A. Abitino, T. L. Hayes, and C. Kanan, "Measuring catastrophic forgetting in neural networks," in *AAAI*. AAAI Press, 2018, pp. 3390–3398.
- [48] K. Liu, B. Dolan-Gavitt, and S. Garg, "Fine-pruning: Defending against backdooring attacks on deep neural networks," in *RAID*, 2018.
- [49] X. Qiao, Y. Yang, and H. Li, "Defending neural backdoors via generative distribution modeling," in *NeurIPS*, 2019, pp. 14004–14013.
- [50] B. Wang, Y. Yao, S. Shan, H. Li, B. Viswanath, H. Zheng, and B. Y. Zhao, "Neural cleanse: Identifying and mitigating backdoor attacks in neural networks," in *2019 IEEE Symposium on Security and Privacy, SP 2019, San Francisco, CA, USA, May 19-23, 2019*. IEEE, 2019.